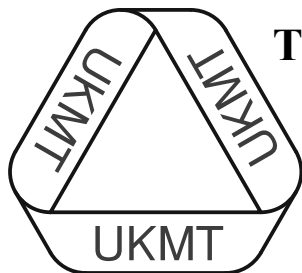


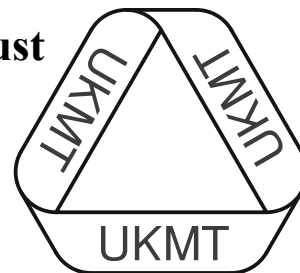
UK Maths Trust

Maclaurin Olympiad

**Past Papers & Solutions
2010-2014**



The United Kingdom Mathematics Trust



**Intermediate Mathematical Olympiad and Kangaroo
(IMOK)**

Olympiad Maclaurin Paper

Thursday 18th March 2010

All candidates must be in *School Year 11* (England and Wales), *S4* (Scotland), or *School Year 12* (Northern Ireland).

READ THESE INSTRUCTIONS CAREFULLY BEFORE STARTING

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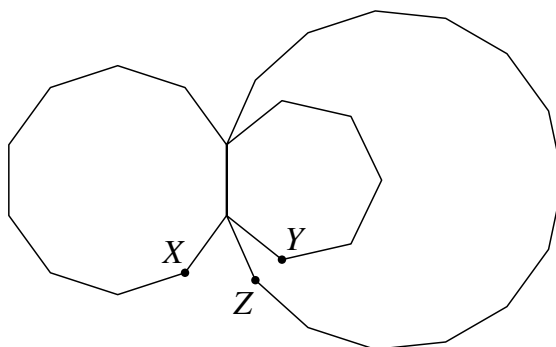
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- How many different ways are there to express $\frac{2}{15}$ in the form $\frac{1}{a} + \frac{1}{b}$, where a and b are positive integers with $a \leq b$?

- The diagram shows a regular heptagon, a regular decagon and a regular 15-gon with an edge in common.
Find the size of angle XYZ .



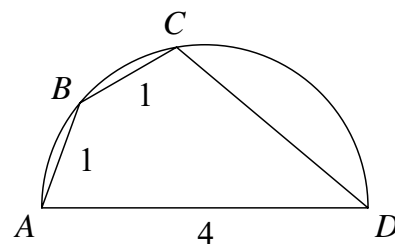
- Solve the equations

$$x + xy + x^2 = 9$$

$$y + xy + y^2 = -3.$$

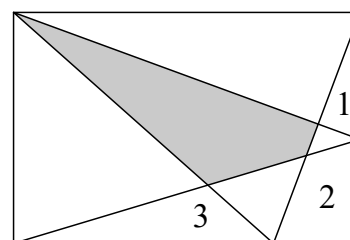
- The diameter AD of a circle has length 4. The points B and C lie on the circle, as shown, so that $AB = BC = 1$.

Find the length of CD .



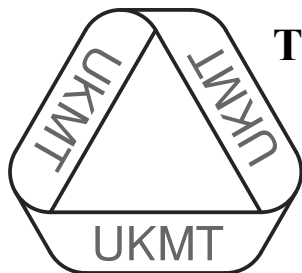
- The diagram shows a rectangle divided into eight regions by four straight lines. Three of the regions have areas 1, 2 and 3, as shown.

What is the area of the shaded quadrilateral?

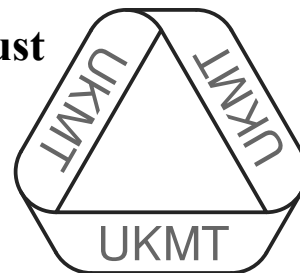


- Every day for the next eleven days I shall eat exactly one sandwich for lunch, either a ham sandwich or a cheese sandwich. However, during that period I shall never eat a ham sandwich on two consecutive days.

In how many ways can I plan my sandwiches for the next eleven days?



The United Kingdom Mathematics Trust



**Intermediate Mathematical Olympiad and Kangaroo
(IMOK)**

Olympiad Maclaurin Paper

Thursday 17th March 2011

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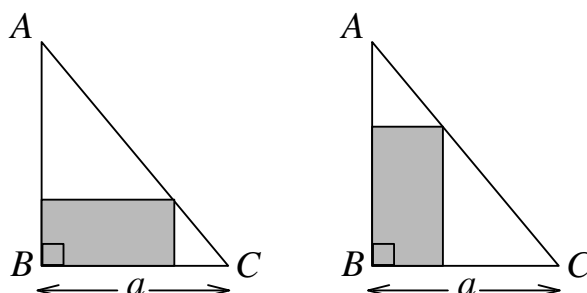
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<http://www.ukmt.org.uk>

- How many positive integers leave a remainder of 31 when divided into 2011?
- I have 44 socks in my drawer, each either red or black. In the dark I randomly pick two socks, and the probability that they do not match is $\frac{192}{473}$.
How many of the 44 socks are red?

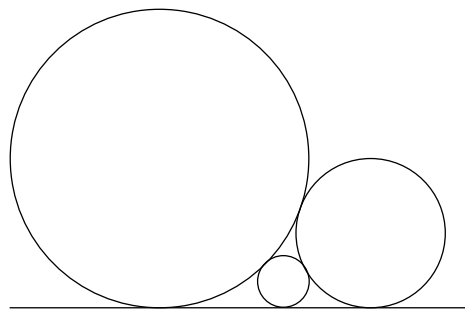
- The diagrams show a rectangle that just fits inside right-angled triangle ABC in two different ways. One side of the triangle has length a .
Prove that the perimeter of the rectangle is $2a$.



- How many solutions are there to the equation $x^2 + y^2 = x^3$, where x and y are positive integers and x is less than 2011?

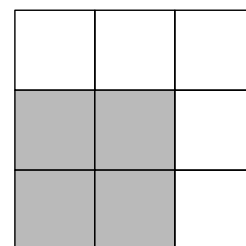
- Three circles touch the same straight line and touch each other, as shown.
Prove that the radii a , b and c , where c is smallest, satisfy the equation

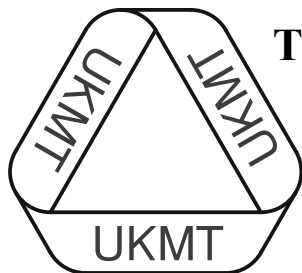
$$\frac{1}{\sqrt{a}} + \frac{1}{\sqrt{b}} = \frac{1}{\sqrt{c}}.$$



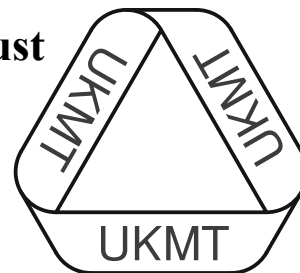
- The numbers 1 to 9 are placed in the cells of a 3×3 square grid, one to each cell. In each of the four 2×2 blocks of adjacent cells, such as the one shaded, the four numbers have the same total T .

What is the maximum possible value of T ?





The United Kingdom Mathematics Trust



**Intermediate Mathematical Olympiad and Kangaroo
(IMOK)**

Olympiad Maclaurin Paper

Thursday 15th March 2012

All candidates must be in *School Year 11* (England and Wales), *S4* (Scotland), or *School Year 12* (Northern Ireland).

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1. Both the digits x and y are nonzero. The five-digit integer 'xyxyx' is divisible by 3 and the seven-digit integer 'xyxyxy' is divisible by 18.

Find all possible values of x and y .

2. Points T and U lie outside parallelogram $PQRS$, and are such that triangles RQT and SRU are equilateral and lie wholly outside the parallelogram.

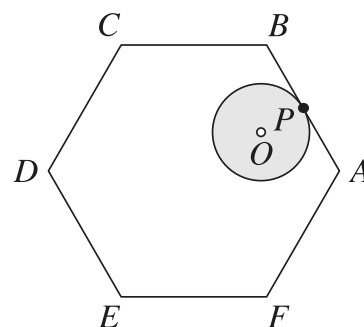
Prove that triangle PTU is equilateral.

3. On Utopia Farm, Farmer Giles has a field in which the amount of grass always increases by the same amount each day. Six cows would take three whole days to eat all the grass in the field; three cows would take seven whole days to eat all the grass in the field.

Assuming that each of Farmer Giles' cows eats the same amount of grass per day, how long would one cow take to eat all the grass in the field?

4. Find all real values of x that satisfy the equation $(1 + x)^4 - 2(1 - x)^4 = (1 - x^2)^2$.

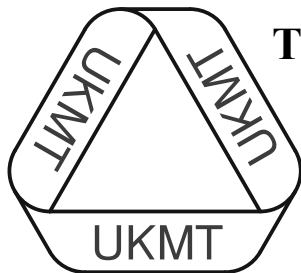
5. A circle with centre O and radius 3 cm rolls without slipping around the inside of a regular hexagon $ABCDEF$. The circle starts with a point P on its circumference in contact with the midpoint of the side AB , as shown. The circle then turns through one complete revolution anticlockwise, touching only sides AB and AF in the process. The circle ends in contact with the side AF , and O , P and A are in a straight line.



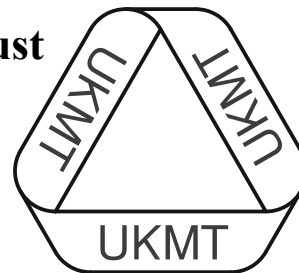
Find the length of AB .

6. Three different positive integers have the property that each of them divides the sum of the other two.

Find all such sets of three numbers.



The United Kingdom Mathematics Trust



**Intermediate Mathematical Olympiad and Kangaroo
(IMOK)**

Olympiad Maclaurin Paper

Thursday 14th March 2013

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<http://www.ukmt.org.uk>

1. Positive integers a , b , c and d are such that

$$a + \frac{1}{b + \frac{1}{c + \frac{1}{d}}} = \frac{20}{13}.$$

What are the values of a , b , c and d ?

2. The positive integer n is such that $n^2 + 20$ is exactly divisible by $n + 2$.

What are the possible values of n ?

3. A circle has centre O and diameter AB . The points C and D lie on the arc $ACDB$ so that AD bisects $\angle BAC$. The line AC is extended to E so that $DE = DA$.

Prove that $\angle CDE$ is a right angle.

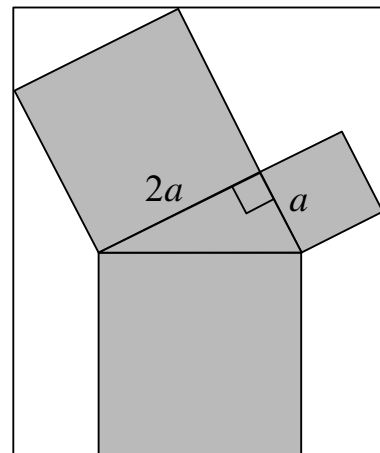
4. Two coins are biased in such a way that, when they are both tossed once:

- (i) the probability of getting two heads is the same as the probability of getting two tails;
- (ii) the probability of getting one head and one tail is $\frac{5}{8}$.

For each coin, what is the probability of getting a head?

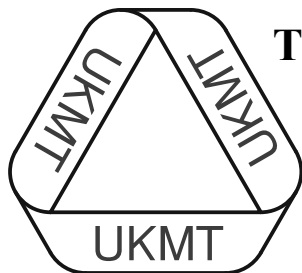
5. Three squares are drawn on the outside of a right-angled triangle, whose shorter sides have lengths a and $2a$. The whole figure is then surrounded by a rectangle, as shown.

What is the ratio of the area of the shaded region to the area of the rectangle?

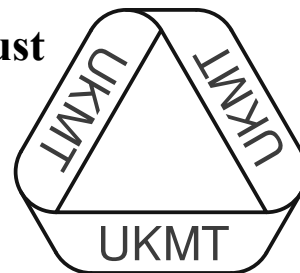


6. All the faces of three identical cubes are to be painted so that 9 faces are green and 9 are yellow. After the paint has dried, the cubes are going to be placed in a velvet bag and given as a gift. So only the colouring can be used to tell two cubes apart.

How many different gifts are possible?



The United Kingdom Mathematics Trust



**Intermediate Mathematical Olympiad and Kangaroo
(IMOK)**

Olympiad Maclaurin Paper

Thursday 20th March 2014

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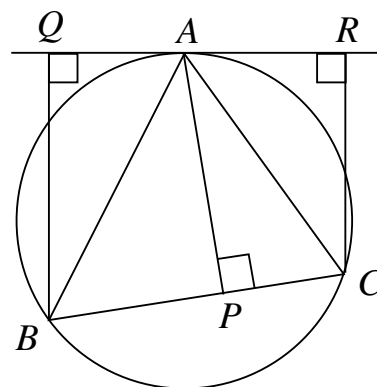
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<http://www.ukmt.org.uk>

1. What is the largest three-digit prime number ' abc ' whose digits a , b and c are different prime numbers?
2. Nine buns cost $\pounds 11 + a$ pence and 13 buns cost $\pounds 15 + b$ pence, where $0 < a < 100$ and $0 < b < 100$.
What is the cost of a bun?
3. A regular hexagon, with sides of length 2 cm, is cut into two pieces by a straight line parallel to one of its sides. The ratio of the area of the smaller piece to the area of the larger piece is 1 : 5.
What is the length of the cut?

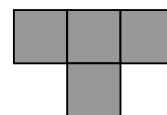
4. In the diagram, RAQ is the tangent at A to the circle ABC , and $\angle AQB$, $\angle CRA$ and $\angle APC$ are all right angles.

Prove that $BQ \times CR = AP^2$.



5. Kim and Oli played nine games of chess, playing alternately with the white and black pieces. Exactly five games were won by whoever was playing with the black pieces, Kim won exactly six games, and no game was drawn.
With which colour pieces did Kim play the first game?

6. The T-tetromino T is the shape made by joining four 1×1 squares edge to edge, as shown. The rectangle R has dimensions $2a \times 2b$, where a and b are integers. The expression ' R can be tiled by T ' means that R can be covered exactly with identical copies of T without gaps or overlaps.



- (a) Prove that R can be tiled by T when both a and b are even.
- (b) Prove that R cannot be tiled by T when both a and b are odd.

Solutions to the 2010 Olympiad Maclaurin Paper

- 1 How many different ways are there to express $\frac{2}{15}$ in the form $\frac{1}{a} + \frac{1}{b}$, where a and b are positive integers with $a \leq b$?

Solution

There are infinitely many integers and hence the number of potential values of a and b is infinite, so any satisfactory method first needs to reduce the problem to a finite number of cases. We demonstrate two methods of doing this.

Method 1

Since $0 < a \leq b$ we have $\frac{1}{b} \leq \frac{1}{a}$ and therefore $\frac{2}{15} = \frac{1}{a} + \frac{1}{b} \leq \frac{2}{a}$. Hence $a \leq 15$.

Also, since $b > 0$, we have $\frac{1}{b} > 0$ and so $\frac{2}{15} = \frac{1}{a} + \frac{1}{b} > \frac{1}{a}$. Hence $a > \frac{15}{2}$.

Therefore $8 \leq a \leq 15$ and the possible values of a are 8, 9, ..., 15. In order to see which of these correspond to integer values of b , it is helpful to find b in terms of a :

$$\begin{aligned} \frac{1}{b} &= \frac{2}{15} - \frac{1}{a} \\ &= \frac{2a - 15}{15a} \end{aligned}$$

and so

$$b = \frac{15a}{2a - 15}.$$

We now see that we require $2a - 15$ to be a positive divisor of $15a$. So we can determine which values of a will give integer values of b from the table:

a	$2a - 15$	$15a$	Divisor?	b
8	1	15×8	yes	120
9	3	15×9	yes	45
10	5	15×10	yes	30
11	7	15×11	no	—
12	9	15×12	yes	20
13	11	15×13	no	—
14	13	15×14	no	—
15	15	15×15	yes	15

Hence there are five ways to express $\frac{2}{15}$ in the required form.

Method 2

We may multiply every term in the equation $\frac{2}{15} = \frac{1}{a} + \frac{1}{b}$ by $15ab$ in order to clear the fractions. We obtain $2ab = 15b + 15a$. We now rearrange this equation, first writing it in the form

$$4ab - 30a - 30b = 0,$$

then adding 225 to both sides to give

$$4ab - 30a - 30b + 225 = 225,$$

that is,

$$(2a - 15)(2b - 15) = 225.$$

Therefore $2a - 15$ is a divisor of $225 = 3^2 \times 5^2$, so that $2a - 15 = 1, 3, 5, 9, 15$. Larger values are not possible since $a \leq b$ and so $2a - 15 \leq 2b - 15$. Also, negative values are not possible since then $2a - 15$ would be at most -15 , but $a > 0$ so that $2a - 15 > -15$.

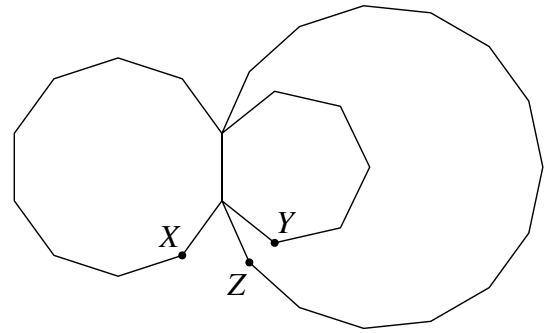
Each of the corresponding values of a and b is an integer:

$$(a, b) = (8, 120), (9, 45), (10, 30), (12, 20) \text{ or } (15, 15).$$

Hence there are five ways to express $\frac{2}{15}$ in the required form.

- 2 The diagram shows a regular heptagon, a regular decagon and a regular 15-gon with an edge in common.

Find the size of angle XYZ .



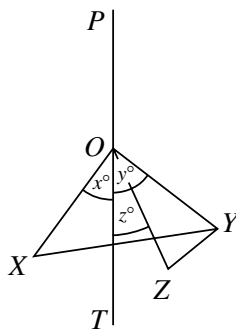
Solution

We shall give two methods, each of which involves finding the exterior angle of a regular polygon. If a regular polygon has n sides, then each exterior angle is given by

$$\text{exterior angle} = \frac{360^\circ}{n}. \quad (2.1)$$

Method 1

Extend the common edge PO to point T as shown below. The angles labelled x° , y° and z° are exterior angles of the regular decagon, regular heptagon and regular 15-gon respectively.



Hence, using the result (2.1),

$$x = \frac{360}{10} = 36,$$

$$y = \frac{360}{7} = 51\frac{3}{7}$$

$$\text{and } z = \frac{360}{15} = 24.$$

It follows that $\angle XOY = x^\circ + y^\circ = 87\frac{3}{7}^\circ$ and $\angle ZOY = y^\circ - z^\circ = 27\frac{3}{7}^\circ$. Now the sides of the three polygons are all equal, so the triangles XOY and ZOY are isosceles. We can therefore find their base angles:

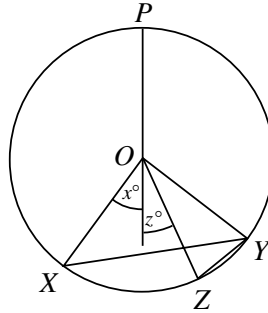
$$\angle XYO = \frac{180^\circ - 87\frac{3}{7}^\circ}{2} = 46\frac{2}{7}^\circ$$

$$\text{and } \angle ZYO = \frac{180^\circ - 27\frac{3}{7}^\circ}{2} = 76\frac{2}{7}^\circ.$$

$$\text{Thus } \angle XYZ = \angle ZYO - \angle XYO = 76\frac{2}{7}^\circ - 46\frac{2}{7}^\circ = 30^\circ.$$

Method 2

Let PO be the common edge, as shown below. Since $OX = OY = OZ = OP$ the points X, Y, Z and P lie on a circle centre O .



Now ‘the angle at the circumference is half the angle at the centre’, so

$$\begin{aligned}\angle XYZ &= \frac{1}{2}\angle XOZ \\ &= \frac{1}{2}(x + z)^\circ.\end{aligned}$$

But from (2.1)

$$x = \frac{360}{10} = 36$$

$$\text{and } z = \frac{360}{15} = 24,$$

$$\text{so that } \angle XYZ = \frac{1}{2}(36 + 24)^\circ = 30^\circ.$$

3 Solve the equations

$$x + xy + x^2 = 9$$

$$y + xy + y^2 = -3.$$

Solution

‘Substitution’ is one of the standard methods of solving simultaneous equations: use one equation to find an expression for one unknown, then substitute this expression into the other equation, thereby forming a single equation in just one of the unknowns. Though it is possible to use a substitution method straight away here, the algebra is rather unpleasant, so we demonstrate two other approaches.

In passing, we note that the question places no restrictions on x and y . In particular, we cannot assume that they are integers.

Method 1

Adding the two given equations, we get

$$x + y + x^2 + 2xy + y^2 = 6$$

so that

$$(x + y)^2 + (x + y) - 6 = 0,$$

which factorises to give

$$(x + y - 2)(x + y + 3) = 0.$$

Hence

$$x + y = 2 \quad \text{or} \quad x + y = -3. \quad (3.1)$$

Also, subtracting the two given equations, we get

$$x - y + x^2 - y^2 = 12$$

which factorises to give

$$(x - y) + (x - y)(x + y) = 12$$

so that

$$(x - y)(1 + x + y) = 12.$$

Hence, using (3.1),

$$x - y = 4 \quad \text{or} \quad x - y = -6, \quad (3.2)$$

which occur when $x + y = 2$ and $x + y = -3$ respectively.

We can now solve equations (3.1) and (3.2) by, for example, adding and subtracting, to obtain

$$(x, y) = (3, -1) \text{ or } \left(-\frac{9}{2}, \frac{3}{2}\right).$$

We now need to check whether these two pairs of values really do satisfy the equations given in the question. Each of them does, so the required solutions are $x = 3, y = -1$ and $x = -\frac{9}{2}, y = \frac{3}{2}$.

Method 2

Factorise each of the given equations to give

$$x(1 + y + x) = 9 \quad (3.3)$$

$$\text{and} \quad y(1 + x + y) = -3. \quad (3.4)$$

Since no side of either equation is zero, we may divide equation (3.3) by (3.4) to obtain $\frac{x}{y} = -3$, so that $x = -3y$. Now substitute this expression for x into equation (3.4) to get

$$y(1 - 2y) = -3,$$

which may be rearranged to

$$2y^2 - y - 3 = 0,$$

or

$$(y + 1)(2y - 3) = 0.$$

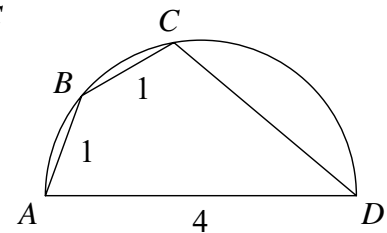
Therefore $y = -1$ or $y = \frac{3}{2}$ and, since $x = -3y$, we have solutions

$$(x, y) = (3, -1) \text{ or } \left(-\frac{9}{2}, \frac{3}{2}\right).$$

We now need to check whether these two pairs of values really do satisfy the equations given in the question. Each of them does, so the required solutions are $x = 3, y = -1$ and $x = -\frac{9}{2}, y = \frac{3}{2}$.

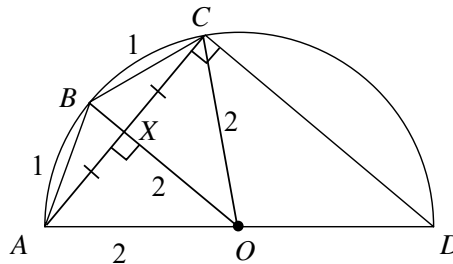
- 4 The diameter AD of a circle has length 4. The points B and C lie on the circle, as shown, so that $AB = BC = 1$.

Find the length of CD .



Solution 1

Let O be the centre of the circle, so that $OA = OB = OC = 2$, and let chord AC and radius OB meet at X .



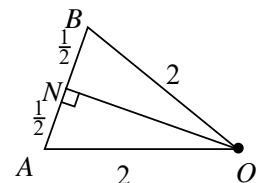
Triangle OAC is isosceles and OB bisects angle AOC (because the chords AB and BC are equal and so subtend equal angles at the centre). Hence OB is the perpendicular bisector of the base AC of the isosceles triangle AOC . In other words, $AX = XC$ and $\angle AXO = 90^\circ$, as shown in the diagram above.

Since angle ACD is 90° (angle in a semicircle) triangle ACD is a right-angled triangle with CD , whose length we have to find, as one side. We know $AD = 4$, so can find CD , using Pythagoras' theorem, from

$$CD^2 = AD^2 - AC^2, \quad (4.1)$$

provided we can find the length of AC . We shall do this by using areas, but there are other methods.

Now consider isosceles triangle OAB and let N be the midpoint of AB , so that triangle ANO is right-angled.



Then, from Pythagoras' theorem, $NO^2 = 2^2 - (\frac{1}{2})^2 = \frac{15}{4}$, so that $NO = \frac{\sqrt{15}}{2}$. Hence the area of triangle AOB is $\frac{1}{2} \times AB \times NO = \frac{\sqrt{15}}{4}$.

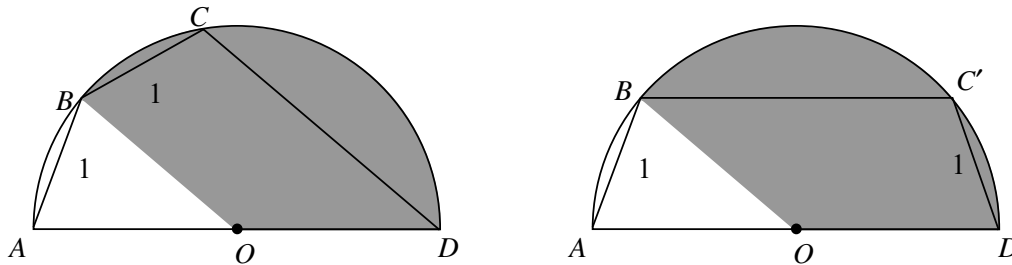
But the area of triangle OAB is also $\frac{1}{2} \times OB \times AX$. Therefore $AX = \frac{\sqrt{15}}{4}$ and so $AC = 2AX = \frac{\sqrt{15}}{2}$. Using this value in equation (4.1), we get

$$CD^2 = AD^2 - AC^2 = 4^2 - \frac{15}{4} = \frac{49}{4}$$

and hence $CD = \frac{7}{2}$.

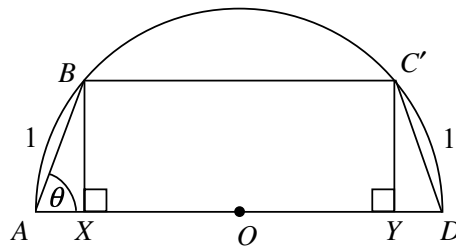
Solution 2

Let O be the centre of the circle. Reflecting the sector OBD , shown shaded, about the diameter perpendicular to BD , so that B and D are interchanged, gives the right-hand diagram. If C' is the reflection of C , then $BC' = CD$. Also, since reflection about a diameter reflects a circle to itself, we know that C' lies on the circle.

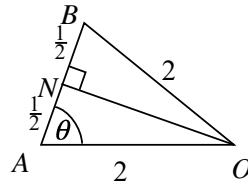


Now angles $AC'B$ and $C'AD$ are both angles at the circumference subtended by chords of length 1. These angles are therefore equal and so BC' and AD are parallel (alternate angles).

Draw perpendiculars from B and C' to AD to create a rectangle $BXYC'$, as shown below, with $BC' = XY$.



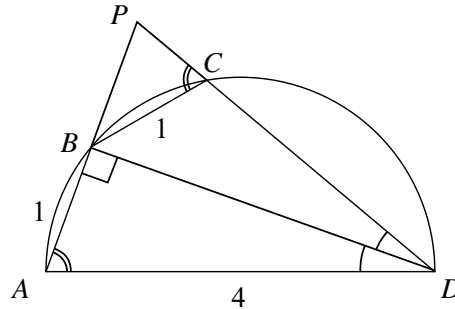
Then $AX = AB \cos \theta = \cos \theta$ and, by constructing the perpendicular bisector of AB in isosceles triangle OAB , shown below, we see that $\cos \theta = \frac{1}{4}$. Therefore $AX = \frac{1}{4}$.



Similarly $DY = \frac{1}{4}$ and so $CD = BC' = XY = 4 - \frac{1}{4} - \frac{1}{4} = 3\frac{1}{2}$.

Solution 3

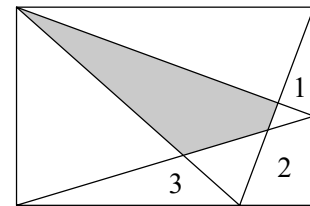
Let P be the intersection of AB produced and DC produced.



Now angles ADB and BDC are both angles at the circumference subtended by chords of length 1. These angles are therefore equal. Also, $\angle ABD = 90^\circ$ (angle in a semicircle). Therefore triangles ABD and PBD are congruent (ASA). Hence $BP = 1$ and $PD = 4$.

Further, in triangles BCP and DAP , angle P is common and $\angle BCP = \angle DAP$ (exterior angle of cyclic quadrilateral). These triangles are therefore similar and hence $PC : 1 = 2 : 4$. So $PC = \frac{1}{2}$ and $CD = PD - PC = 4 - \frac{1}{2} = 3\frac{1}{2}$.

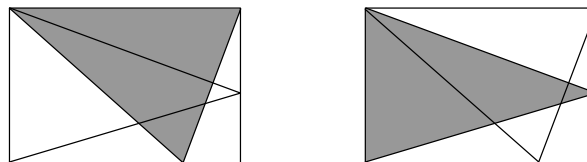
- 5 The diagram shows a rectangle divided into eight regions by four straight lines. Three of the regions have areas 1, 2 and 3, as shown.



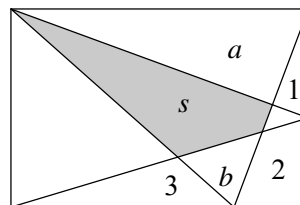
What is the area of the shaded quadrilateral?

Solution

Notice that the shaded area is the intersection of the two triangles shown shaded in the following diagrams.



The area of each of these triangles is half the area of the rectangle, since each has the same base and height as the rectangle. Therefore the total unshaded area within the rectangle in the right-hand figure is also half the area of the rectangle. In other words, the shaded area in the left-hand figure is equal to the total unshaded area in the right-hand figure.



Using the notation indicated above, we therefore have $a + s + b = a + 1 + 2 + b + 3$ and hence $s = 1 + 2 + 3 = 6$. So the area of the shaded quadrilateral is 6.

Remark

The result also applies if the rectangle is replaced by a parallelogram. Moreover, there is clearly nothing special about the values 1, 2 and 3.

- 6 Every day for the next eleven days I shall eat exactly one sandwich for lunch, either a ham sandwich or a cheese sandwich. However, during that period I shall never eat a ham sandwich on two consecutive days.

In how many ways can I plan my sandwiches for the next eleven days?

Solution

We shall use the notation $\binom{n}{r}$ for the number of ways of choosing r objects from n . This is a binomial coefficient, sometimes written nC_r . We have

$$\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)\dots(n-r+1)}{1 \times 2 \times \dots \times r}.$$

We give three different methods, all of which create a plan for the next eleven days by constructing a line of Cs and Hs.

Method 1

Let $h(n)$ be the number of ways of creating a line of length n starting with H, and $c(n)$ be the number starting with C.

For any line of length n starting with C, we may construct a line of length $n+1$ starting with H by adding H at the start. Also, for any line of length $n+1$ starting with H, we may construct a line of length n starting with C by deleting the initial H. Both of these statements depend on the rule that there are no consecutive Hs. Hence

$$h(n+1) = c(n).$$

Similarly, adding C to the front of a line starting with H or one starting with C creates a longer line starting with C (and *vice versa*), so that $c(n+1) = c(n) + h(n)$.

Therefore

$$\begin{aligned} h(n+2) &= c(n+1) \\ &= c(n) + h(n) \\ &= h(n+1) + h(n) \end{aligned}$$

and

$$\begin{aligned} c(n+2) &= c(n+1) + h(n+1) \\ &= c(n+1) + c(n). \end{aligned}$$

We thus have two sequences generated like the Fibonacci sequence: each term is the sum of the previous two terms. Moreover, letting $t(n)$ be the total number of ways to construct a line for n days, then $t(n) = h(n) + c(n)$ and therefore $t(n+2) = t(n+1) + t(n)$. Hence $t(n)$ is also a Fibonacci-like sequence.

Now $t(1) = 2$, corresponding to the 'lines' C and H, and $t(2) = 3$, corresponding to the lines CC, CH and HC. So the first eleven terms of $t(n)$ are 2, 3, 5, 8, 13, 21, 34, 55, 89, 144 and 233. Hence the number required is 233.

Method 2

We create a plan for the next eleven days by constructing a line of Cs and Hs from two types of tile, $\boxed{C}$ or $\boxed{HC}$, which ensures that two Hs are never placed together. There are various possibilities, determined by the number of $\boxed{C}$ tiles which are used, which can be 11, 9, 7, 5, 3 or 1.

Eleven $\boxed{C}$ tiles can be placed in just one way, or $\binom{11}{0}$ ways.

Nine $\boxed{C}$ tiles and one $\boxed{HC}$ tile can be placed in $\binom{10}{1}$ ways, the number of ways of choosing the position of one $\boxed{HC}$ tile in a row of 10 tiles.

Seven $\boxed{C}$ tiles and two $\boxed{HC}$ tiles can be placed in $\binom{9}{2}$ ways, the number of ways of choosing the position of two $\boxed{HC}$ tiles in a row of 9 tiles.

Continuing in this way, we see that the total number of ways is

$$\begin{aligned} \binom{11}{0} + \binom{10}{1} + \binom{9}{2} + \binom{8}{3} + \binom{7}{4} + \binom{6}{5} &= 1 + 10 + 36 + 56 + 35 + 6 \\ &= 144. \end{aligned}$$

However, placing tiles in this way always ends the line with a C, so does not allow for the possibility of ending with an H. But if the line ends with an H, then the remainder of the line has ten letters and ends with a C. We may count the number of ways for this in a similar way to the above, but with a total of 10 letters instead of 11:

$$\begin{aligned} \binom{10}{0} + \binom{9}{1} + \binom{8}{2} + \binom{7}{3} + \binom{6}{4} + \binom{5}{5} &= 1 + 9 + 28 + 35 + 15 + 1 \\ &= 89. \end{aligned}$$

Therefore the total number of ways altogether is $144 + 89 = 233$.

Method 3

We create a plan for the next eleven days by starting with k Cs in a line, where $k \leq 11$, and adding some Hs (possibly none) to construct a line of Cs and Hs. There are $k + 1$ slots into which a single H may be placed—at either end or between two Cs. We need to place $(11 - k)$ Hs so that each H is in a *different* slot, so we have to choose $11 - k$ of the $k + 1$ slots; there are $\binom{k + 1}{11 - k}$ ways in which this can be done.

Now we cannot add more Hs than the number of available slots, so $11 - k \leq k + 1$, that is, $k \geq 5$.

Therefore the total number of ways is the sum of $\binom{k + 1}{11 - k}$ for $5 \leq k \leq 11$, in other words

$$\begin{aligned} \binom{6}{6} + \binom{7}{5} + \binom{8}{4} + \binom{9}{3} + \binom{10}{2} + \binom{11}{1} + \binom{12}{0} &= 1 + 21 + 70 + 84 + 45 + 11 + 1 \\ &= 233. \end{aligned}$$

Solutions to the 2011 Olympiad Maclaurin Paper

1. How many positive integers leave a remainder of 31 when divided into 2011?

Solution

If an integer leaves a remainder of 31 when divided into 2011, then it divides exactly into $2011 - 31 = 1980$. Also, in order to leave a remainder of 31, the integer itself needs to be greater than 31.

Now $1980 = 2^2 \times 3^2 \times 5 \times 11$. Hence any divisor of 1980 may be obtained by choosing one term from each of the following four lists and multiplying them together:

$$1, 2, 2^2;$$

$$1, 3, 3^2;$$

$$1, 5;$$

$$1, 11.$$

There are $3 \times 3 \times 2 \times 2 = 36$ ways of choosing the terms and therefore 1980 has 36 divisors.

The divisors of 1980 which are less than or equal to 31 are

$$1, 2, 3, 4, 5, 6, 9, 10, 11, 12, 15, 18, 20, 22 \text{ and } 30,$$

that is, 15 divisors in all. Hence the number of divisors of 1980 which are greater than 31 is $36 - 15 = 21$.

Therefore there are 21 positive integers which leave a remainder of 31 when divided into 2011.

2. I have 44 socks in my drawer, each either red or black. In the dark I randomly pick two socks, and the probability that they do not match is $\frac{192}{473}$.

How many of the 44 socks are red?

Solution

Let there be r red socks and so there are $44 - r$ black socks.

To pick a non-matching pair, I would either have to choose a red followed by a black or vice-versa. The probability of choosing a red and then a black is

$$\frac{r}{44} \times \frac{44 - r}{43}$$

and the probability of choosing a black and then a red is

$$\frac{44 - r}{44} \times \frac{r}{43}.$$

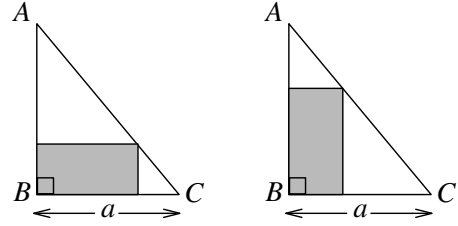
These two probabilities are identical, and hence the probability of picking a non-matching pair is

$$\frac{2r(44 - r)}{44 \times 43} = \frac{192}{473}.$$

This equation simplifies to $r(44 - r) = 384$, that is, $r^2 - 44r + 384 = 0$. After factorising, we obtain $(r - 12)(r - 32) = 0$ and so $r = 12$ or $r = 32$. In other words, there are either 12 red socks or 32 red socks.

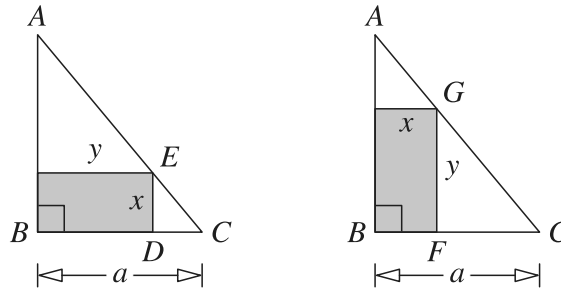
Notice that these two solutions are symmetrical; I must have 12 socks of one colour and 32 of the other.

3. The diagrams show a rectangle that just fits inside right-angled triangle ABC in two different ways. One side of the triangle has length a . Prove that the perimeter of the rectangle is $2a$.



Solution

We add some labels to the diagrams and let the dimensions of the rectangle be $x \times y$, as shown.



The triangles DCE and FCG are similar, so

$$\frac{x}{a - y} = \frac{y}{a - x},$$

which rearranges to give

$$x(a - x) = y(a - y)$$

that is,

$$\begin{aligned} a(x - y) &= x^2 - y^2 \\ &= (x - y)(x + y). \end{aligned}$$

Since $x - y \neq 0$ (because there are two *different* ways of fitting the rectangle) this implies that $a = x + y$. Thus the perimeter of the rectangle is $2x + 2y = 2a$.

Note

It follows that the triangle ABC is actually isosceles. Can you see why?

4. How many solutions are there to the equation $x^2 + y^2 = x^3$, where x and y are positive integers and x is less than 2011?

Solution

If $x^2 + y^2 = x^3$, then $y^2 = x^2(x - 1)$ and therefore, since $x > 0$,

$$\frac{y^2}{x^2} = x - 1. \quad (4.1)$$

We know that $x - 1$ is an integer. We deduce from equation 4.1 that $x - 1$ is a perfect square. Let $x - 1 = n^2$ for some non-negative integer n .

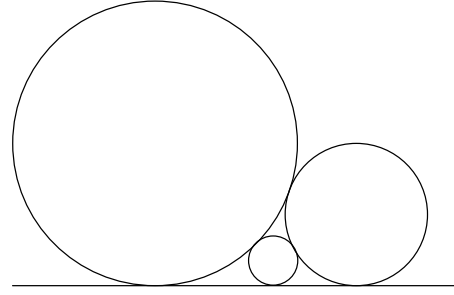
Now $0 < x < 2011$, so that $-1 < n^2 < 2010$ and hence $0 \leq n \leq 44$. But when $n = 0$ we have $x = 1$ and so $y = 0$, which is not allowed. Each other value of n gives a unique value of x , and therefore of y since $y > 0$.

Hence there are 44 solutions to the given equation in positive integers.

5. Three circles touch the same straight line and touch each other, as shown.

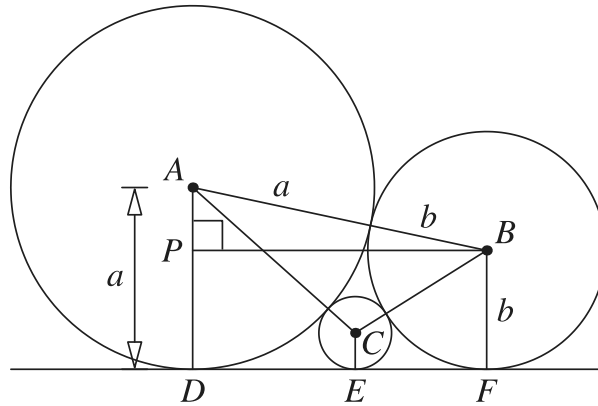
Prove that the radii a , b and c , where c is smallest, satisfy the equation

$$\frac{1}{\sqrt{a}} + \frac{1}{\sqrt{b}} = \frac{1}{\sqrt{c}}.$$



Solution

We add some labels and lines to the diagram, as shown below: A , B and C are the centres; D , E and F are the points of contact of the circles and the straight line; and P is the foot of the perpendicular from B to AD .



Note first that AB , BC and CA , the lines of centres, pass through the points of tangency of the pairs of circles. Also, AD , CE and BF are perpendicular to the line DEF .

Then, by Pythagoras' theorem applied to triangle APB ,

$$(a - b)^2 + PB^2 = (a + b)^2.$$

Therefore

$$\begin{aligned} PB^2 &= (a + b)^2 - (a - b)^2 \\ &= 4ab. \end{aligned}$$

Hence $DF = PB = 2\sqrt{ab}$. Similarly $DE = 2\sqrt{ca}$ and $EF = 2\sqrt{bc}$.

Since $DF = DE + EF$ we have

$$2\sqrt{ab} = 2\sqrt{ca} + 2\sqrt{bc}.$$

Dividing through by $2\sqrt{abc}$ gives the required result.

6. The numbers 1 to 9 are placed in the cells of a 3×3 square grid, one to each cell. In each of the four 2×2 blocks of adjacent cells, such as the one shaded, the four numbers have the same total T .

What is the maximum possible value of T ?

Solution

The total of the numbers in the grid is $1 + 2 + \dots + 9 = 45$.

Let the central number be x , let the four middle numbers along the sides be a, b, c, d and let the four corner numbers be p, q, r, s , as shown in the diagram:

p	a	q
b	x	c
r	d	s

Now add up the totals of the numbers in the four 2×2 blocks in two ways to show that

$$\begin{aligned} 4T &= 4x + 2(a + b + c + d) + (p + q + r + s) \\ &= 45 + 3x + (a + b + c + d). \end{aligned}$$

Since $x \leq 9$ and $x + a + b + c + d \leq 9 + 8 + 7 + 6 + 5 = 35$ we have $4T \leq 45 + 18 + 35 = 98$. Therefore $T \leq 24$.

However, it is possible to achieve the value 24 in several ways, for instance:

4	3	5
8	9	7
1	6	2

1	6	3
9	8	7
2	5	4

We deduce that the maximum value of T is certainly 24.

Note

Notice that, somewhat counter-intuitively, it is possible to achieve the maximum T using 8 in the central cell.

Solutions to the 2012 Olympiad Maclaurin Paper

1. Both the digits x and y are nonzero. The five-digit integer ' $xyxyx$ ' is divisible by 3 and the seven-digit integer ' $yxyxyxy$ ' is divisible by 18.

Find all possible values of x and y .

Solution

Since 3 is a factor of ' $xyxyx$ ' we know that 3 is a factor of the digit sum $3x + 2y$ and hence of $2y$. But 2 is coprime to 3, so 3 is a factor of y .

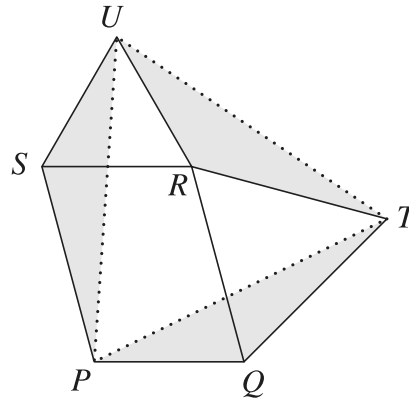
The integer ' $yxyxyxy$ ' is even, so y is even. Hence we deduce that $y = 6$, the only even digit that has 3 as a factor.

Now 9 is a factor of ' $6x6x6x6$ ' and so it is also a factor of the digit sum $24 + 3x$, and hence of $6 + 3x$. Thus $x = 1, 4$ or 7 , the only digits for which this is the case.

2. Points T and U lie outside parallelogram $PQRS$, and are such that triangles RQT and SRU are equilateral and lie wholly outside the parallelogram.

Prove that triangle PTU is equilateral.

Solution



From the properties of a parallelogram and an equilateral triangle we have $PQ = SR = US = UR$ and $SP = RQ = QT = RT$. Also

$$\begin{aligned}\angle USP &= 60^\circ + \angle RSP \\ &= 60^\circ + \angle PQR \\ &= \angle PQT,\end{aligned}$$

and, from 'angles at a point',

$$\begin{aligned}\angle URT &= 360^\circ - 60^\circ - 60^\circ - \angle QRS \\ &= 240^\circ - (180^\circ - \angle PQR) \\ &= 60^\circ + \angle PQR \\ &= \angle PQT.\end{aligned}$$

Therefore triangles USP , PQT and URT are congruent (SAS) and it follows that $UP = PT = TU$.

3. On Utopia Farm, Farmer Giles has a field in which the amount of grass always increases by the same amount each day. Six cows would take three whole days to eat all the grass in the field; three cows would take seven whole days to eat all the grass in the field.

Assuming that each of Farmer Giles' cows eats the same amount of grass per day, how long would one cow take to eat all the grass in the field?

Solution

Let F be the amount of grass in the field initially, C the amount a cow eats in one day and A the increase of grass in one day. Then from the given information we have

$$F + 3A = 6 \times 3 \times C \quad (1)$$

$$\text{and} \quad F + 7A = 3 \times 7 \times C. \quad (2)$$

Subtracting equation (1) from equation (2), we obtain $4A = 3C$ and hence $F = 21A$.

Now let n be the number of days it would take one cow to eat all of the grass. Then we have $F + nA = nC$, and, using the values above, we obtain $n = 63$. Thus it would take 63 days for one cow to eat all the grass.

4. Find all real values of x that satisfy the equation $(1+x)^4 - 2(1-x)^4 = (1-x^2)^2$.

Solution

Let $a = 1 + x$ and $b = 1 - x$. Then $ab = 1 - x^2$ and the given equation becomes $a^4 - 2b^4 = a^2b^2$, so that

$$a^4 - a^2b^2 - 2b^4 = 0,$$

that is,

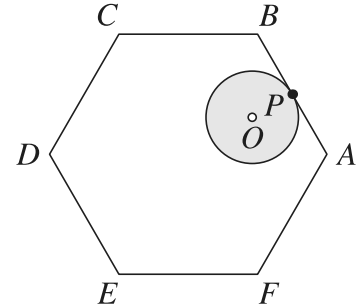
$$(a^2 + b^2)(a^2 - 2b^2) = 0.$$

Hence either $a^2 + b^2 = 0$ or $a^2 - 2b^2 = 0$. In the first case $a = b = 0$, which is impossible; in the second case $a = \pm\sqrt{2}b$. In terms of x , this means that $1+x = \pm\sqrt{2}(1-x)$, so that

$$\text{either} \quad x = \frac{\sqrt{2}-1}{\sqrt{2}+1} = \frac{\sqrt{2}-1}{\sqrt{2}+1} \times \frac{\sqrt{2}-1}{\sqrt{2}-1} = 3 - 2\sqrt{2},$$

$$\text{or} \quad x = \frac{\sqrt{2}+1}{\sqrt{2}-1} = \frac{\sqrt{2}+1}{\sqrt{2}-1} \times \frac{\sqrt{2}+1}{\sqrt{2}+1} = 3 + 2\sqrt{2}.$$

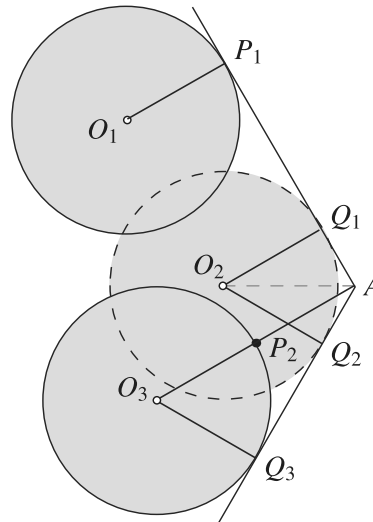
5. A circle with centre O and radius 3 cm rolls without slipping around the inside of a regular hexagon $ABCDEF$. The circle starts with a point P on its circumference in contact with the midpoint of the side AB , as shown. The circle then turns through one complete revolution anticlockwise, touching only sides AB and AF in the process. The circle ends in contact with the side AF , and O , P and A are in a straight line.



Find the length of AB .

Solution

Let O_1 and P_1 be the original positions of O and P . When the disc reaches the corner at A , let O_2 be the position of the centre and let Q_1 and Q_2 be the points of contact with AB and AF . Finally, when the disc has revolved exactly once, let O_3 be the centre, let Q_3 be the point of contact and let P_2 be the final position of P .



Now $\angle Q_1AQ_2 = 120^\circ$ and $O_2Q_1 = 3$, so $AQ_1 = AQ_2 = \sqrt{3}$. Since the circle has completed one revolution without slipping $P_1Q_1 + Q_2Q_3 = 6\pi$.

Also O_3A is perpendicular to AP_1 , so $\angle O_3AQ_3 = 30^\circ$. Hence $AQ_3 = 3\sqrt{3}$. Letting $P_1A = a$ and putting all of this together, we obtain

$$P_1Q_1 + Q_2Q_3 = (a - \sqrt{3}) + (3\sqrt{3} - \sqrt{3}) = 6\pi$$

and so $a = 6\pi - \sqrt{3}$. Therefore the length $2a$ of the side of the hexagon is $12\pi - 2\sqrt{3}$.

6. Three different positive integers have the property that each of them divides the sum of the other two.

Find all such sets of three numbers.

Solution

Let the integers be a, b, c , where $0 < a < b < c$. Hence $a + b < 2c$, but c is a factor of $a + b$, so $a + b = c$.

Thus the integers are a, b and $a + b$.

Now we also require b to divide $2a + b$. It follows that b divides $2a$. But $2a < 2b$ so $b = 2a$, and therefore $c = 3a$. But then the condition that a divides $b + c$ automatically holds.

Hence there is an infinite number of solutions: $(a, b, c) = (k, 2k, 3k)$, where k is any positive integer.

Note: For any $n \geq 3$, there exists a list of n different positive integers such that each of them divides the sum of the remaining $n - 1$ numbers. However, classifying all solutions for general n is an unsolved problem. One possible list is: $1, 2, 3 \times 2^0, 3 \times 2^1, \dots, 3 \times 2^{n-3}$. Can you prove that this list has the desired property?

Solutions to the 2013 Olympiad Maclaurin Paper

1. Positive integers a, b, c and d are such that

$$a + \frac{1}{b + \frac{1}{c + \frac{1}{d}}} = \frac{20}{13}.$$

What are the values of a, b, c and d ?

Solution

Since

$$\frac{20}{13} = 1 + \frac{7}{13}$$

it follows that $a = 1$ and

$$\frac{1}{b + \frac{1}{c + \frac{1}{d}}} = \frac{7}{13}.$$

Therefore

$$\begin{aligned} b + \frac{1}{c + \frac{1}{d}} &= \frac{13}{7} \\ &= 1 + \frac{6}{7} \end{aligned}$$

and so $b = 1$.

Continuing this process, we find that $c = 1$ and $d = 6$.

Note: The expression

$$1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{6}}}$$

is an example of a continued fraction.

2. The positive integer n is such that $n^2 + 20$ is exactly divisible by $n + 2$.

What are the possible values of n ?

Solution

We are given that $n^2 + 20$ is exactly divisible by $n + 2$. Since

$$n^2 + 20 = (n - 2)(n + 2) + 24$$

it follows that 24 is exactly divisible by $n + 2$.

Now the divisors of 24 are $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 8, \pm 12$ and ± 24 . However, n is positive, so $n + 2$ is at least 3.

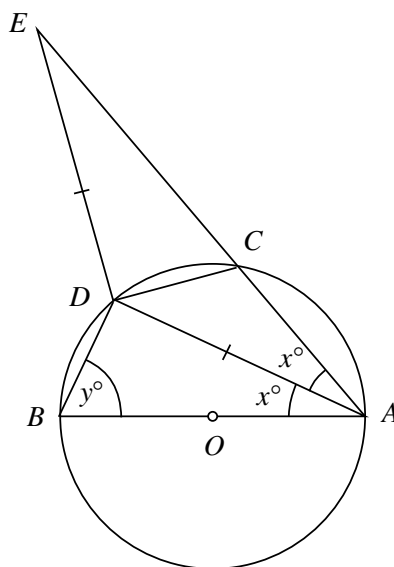
Therefore the possible values of $n + 2$ are 3, 4, 6, 8, 12 and 24, and the corresponding values of n are 1, 2, 4, 6, 10 and 22.

3. A circle has centre O and diameter AB . The points C and D lie on the arc $ACDB$ so that AD bisects $\angle BAC$. The line AC is extended to E so that $DE = DA$.

Prove that $\angle CDE$ is a right angle.

Solution

Let $\angle BAD$ and $\angle DAC$ be x° , and let $\angle DBA$ be y° , as shown in the figure.



By 'angle in a semicircle', we have $\angle ADB = 90^\circ$, so that $x + y = 90$ from the angle sum of triangle ADB .

Since $DE = DA$, from 'base angles of an isosceles triangle' we have $\angle CED = \angle DAC$, so that $\angle CED = x^\circ$.

Also, from 'external angle of a cyclic quadrilateral' we have $\angle DCE = \angle DBA$, so that $\angle DCE = y^\circ$.

Finally, from the angle sum of triangle DCE it follows that

$$\angle CDE = 180^\circ - x^\circ - y^\circ = 180^\circ - 90^\circ = 90^\circ,$$

as required.

4. Two coins are biased in such a way that, when they are both tossed once:

- (i) the probability of getting two heads is the same as the probability of getting two tails;
- (ii) the probability of getting one head and one tail is $\frac{5}{8}$.

For each coin, what is the probability of getting a head?

Solution

Let p be the probability of a head on one coin and let q be the probability of a head on the other coin.

Then from condition (i) we have

$$pq = (1 - p)(1 - q),$$

from which, after simplifying, we obtain

$$p + q = 1. \tag{1}$$

Also, from condition (ii) we have

$$p(1 - q) + q(1 - p) = \frac{5}{8},$$

which may be simplified to

$$p + q - 2pq = \frac{5}{8}.$$

Substituting from equation (1), we find

$$1 - 2pq = \frac{5}{8},$$

from which we obtain

$$pq = \frac{3}{16}. \tag{2}$$

From equations (1) and (2) we deduce that p and q are the roots of the quadratic equation

$$16x^2 - 16x + 3 = 0$$

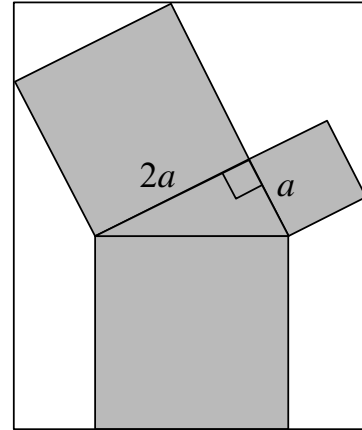
that is

$$(4x - 3)(4x - 1) = 0,$$

so that $x = \frac{3}{4}$ or $x = \frac{1}{4}$. Hence, the two required probabilities are $\frac{1}{4}$ and $\frac{3}{4}$.

5. Three squares are drawn on the outside of a right-angled triangle, whose shorter sides have lengths a and $2a$. The whole figure is then surrounded by a rectangle, as shown.

What is the ratio of the area of the shaded region to the area of the rectangle?



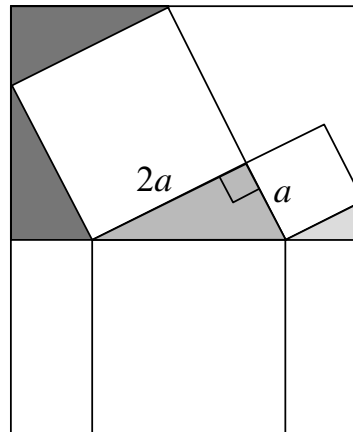
Solution

The areas of the two upper squares are a^2 and $4a^2$, so by Pythagoras' theorem the area of the lower square is $5a^2$, and therefore the sides of the lower square have length $\sqrt{5}a$.

Hence the total shaded area is

$$a^2 + 4a^2 + 5a^2 + \frac{1}{2}a \times 2a = 11a^2.$$

Now the four shaded triangles in the following figure are all similar, because their corresponding angles are equal. (In fact, two of these triangles are congruent.)



But the original right-angled triangle has sides of length a , $2a$ and $\sqrt{5}a$, so each of these similar triangles has side lengths in the ratio $1 : 2 : \sqrt{5}$.

From this, the breadth of the rectangle is

$$\frac{1}{\sqrt{5}} \times 2a + \sqrt{5}a + \frac{2}{\sqrt{5}} \times a = \frac{9\sqrt{5}}{5}a$$

and the height is

$$\sqrt{5}a + \frac{2}{\sqrt{5}} \times 2a + \frac{1}{\sqrt{5}} \times 2a = \frac{11\sqrt{5}}{5}a.$$

Hence the area of the rectangle is

$$\frac{9\sqrt{5}}{5}a \times \frac{11\sqrt{5}}{5}a = \frac{99}{5}a^2.$$

Therefore the ratio of the area of the shaded region to the area of the rectangle is

$$11a^2 : \frac{99}{5}a^2 = 5 : 9.$$

6. All the faces of three identical cubes are to be painted so that 9 faces are green and 9 are yellow. After the paint has dried, the cubes are going to be placed in a velvet bag and given as a gift. So only the colouring can be used to tell two cubes apart.

How many different gifts are possible?

Solution

It is worth first paying attention to the number of different ways in which a cube can be painted using one colour, say green, to paint various numbers of faces, with all the remaining faces of the other colour.

If 0, 1, 5 or 6 faces are painted green, then there is only one way. If two faces are painted green, then these might be opposite or adjacent faces, so there are two possible ways. If three faces are painted green, then the faces might form a 'collar' (with two opposite faces and one other) or a 'cap' (with no opposite faces), so there are two possible ways.

Now we consider the distribution of faces between the three cubes, in each case counting the number of possible gifts. Note that the order of cubes is not important. The numbers in the first three columns are the number of faces painted green; clearly the remaining faces are painted yellow.

Cube A	Cube B	Cube C	Gifts
6	3	0	$1 \times 2 \times 1 = 2$
6	2	1	$1 \times 2 \times 1 = 2$
5	4	0	$1 \times 2 \times 1 = 2$
5	3	1	$1 \times 2 \times 1 = 2$
5	2	2	3
4	4	1	3
4	3	2	$2 \times 2 \times 2 = 8$
3	3	3	4

In the 5, 2, 2 case, there is just one way to colour cube A, and cubes B and C can either be two opposites, two adjacents or one of each, so there are three different gifts. In the 4, 4, 1 case the argument is the same. In the 3, 3, 3 case, the number of collars can be either 0, 1, 2 or 3, with the remainder being caps, so there are four different gifts.

This yields a total of 26 different gifts.

Solutions to the Olympiad Maclaurin Paper

- M1.** What is the largest three-digit prime number ‘ abc ’ whose digits a , b and c are different prime numbers?

Solution

The primes available as digits are 2, 3, 5 and 7.

The three-digit prime p sought cannot end in 2 or 5, since then it would be divisible by 2 or 5. Also, p cannot consist of the digits 2, 3 and 7 or the digits 3, 5 and 7 since it would then be divisible by 3.

Hence p is composed either of 2, 3 and 5 or of 2, 5 and 7.

Now p cannot begin with 7 since it would then end in 2 or 5. The next biggest candidate is 527. But 527 is not a prime because $527 = 17 \times 31$.

However, the next biggest candidate 523 is prime. We can confirm this by showing that 523 has no factors less than 23, which is sufficient since $23^2 = 529 > 523$. Our evidence for this is that

$$\begin{aligned} 523 &= 7 \times 75 - 2 \\ &= 11 \times 47 + 6 \\ &= 13 \times 40 + 3 \\ &= 17 \times 30 + 13 \\ &= 19 \times 27 + 10. \end{aligned}$$

- M2.** Nine buns cost £11 + a pence and 13 buns cost £15 + b pence, where $0 < a < 100$ and $0 < b < 100$.

What is the cost of a bun?

Solution

Let a bun cost x pence.

Since $9x = 1100 + a$ and $0 \leq a < 100$, we have $1100 \leq 9x < 1200$. Therefore $123 \leq x \leq 133$.

Similarly, since $13x = 1500 + b$ and $0 \leq b < 100$, we have $1500 \leq 13x < 1600$. Therefore $116 \leq x \leq 123$.

It follows that $x = 123$ (and $a = 7, b = 99$). Hence the cost of a bun is £1.23.

- M3.** A regular hexagon, with sides of length 2 cm, is cut into two pieces by a straight line parallel to one of its sides. The ratio of the area of the smaller piece to the area of the larger piece is 1 : 5.

What is the length of the cut?

Solution

For convenience, we omit units throughout our working: all lengths are measured in cm, and all areas in cm^2 .

The hexagon may be divided into six equilateral triangles with sides of length 2, as shown in

Figure 1. Each triangle has area $\frac{1}{2} \times 2 \times \sqrt{3} = \sqrt{3}$, so the area of the hexagon is $6\sqrt{3}$.

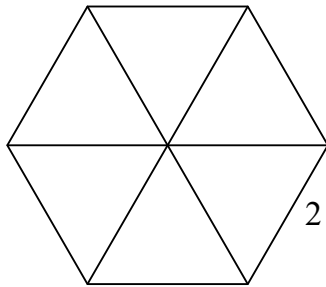


Figure 1

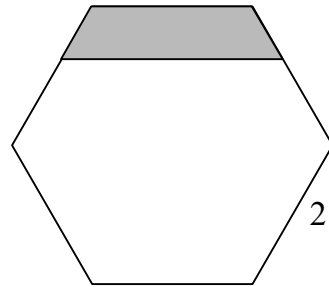
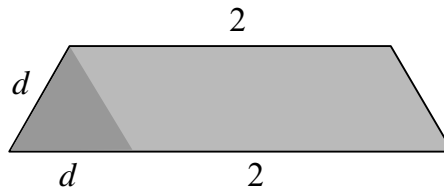


Figure 2

Figure 2 shows the hexagon cut into two pieces by the straight line parallel to a side. The ratio of the areas of the two pieces is 1 : 5, so that the shaded area is equal to $\sqrt{3}$.

Now the shaded region is a trapezium, which we may divide into an equilateral triangle of side d and a parallelogram, as shown below.



The height of the trapezium is equal to the height of the triangle, which is $\frac{\sqrt{3}}{2}d$.

Therefore the area of the trapezium is

$$\frac{1}{2}(2 + 2 + d) \times \frac{\sqrt{3}}{2}d = \frac{\sqrt{3}d(4 + d)}{4}.$$

Thus we have

$$\frac{\sqrt{3}d(4 + d)}{4} = \sqrt{3},$$

that is,

$$4d + d^2 = 4.$$

Hence

$$(2 + d)^2 = 8,$$

so that, because d is positive and therefore $2 + d$ is,

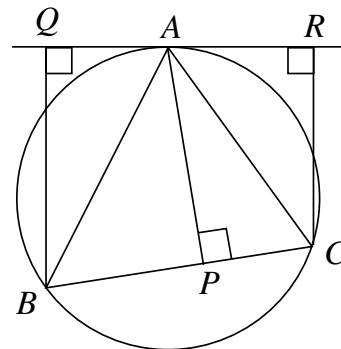
$$2 + d = \sqrt{8}.$$

In other words, the length of the cut is $\sqrt{8}$, which equals $2\sqrt{2}$.

3

- M4.** In the diagram, RAQ is the tangent at A to the circle ABC , and $\angle AQB$, $\angle CRA$ and $\angle APC$ are all right angles.

Prove that $BQ \times CR = AP^2$.



Solution

By the alternate segment theorem, since RAQ is a tangent we have $\angle QAB = \angle BCA$. Also $\angle AQB = 90^\circ = \angle APC$. Hence the triangles QAB and PCA are similar (AA), so that

$$\frac{BQ}{AP} = \frac{BA}{AC}. \quad (1)$$

Furthermore, $\angle RAC = \angle ABC$ (again by the alternate segment theorem) and $\angle ARC = 90^\circ = \angle BPA$. Hence the triangles ARC and BPA are also similar (AA), so that

$$\frac{CR}{AP} = \frac{AC}{BA}. \quad (2)$$

From equations (1) and (2), we obtain

$$\begin{aligned} \frac{BQ}{AP} \times \frac{CR}{AP} &= \frac{BA}{AC} \times \frac{AC}{BA} \\ &= 1 \end{aligned}$$

and so

$$BQ \times CR = AP^2,$$

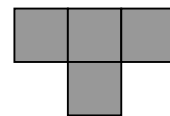
as required.

- M5.** Kim and Oli played nine games of chess, playing alternately with the white and black pieces. Exactly five games were won by whoever was playing with the black pieces, Kim won exactly six games, and no game was drawn.
With which colour pieces did Kim play the first game?

Solution

Suppose that Kim wins b games playing Black. Then Kim wins $6 - b$ playing White. Now Oli wins $5 - b$ games playing Black, therefore Kim loses $5 - b$ games playing White. Hence altogether the number of games when Kim plays White is $(6 - b) + (5 - b) = 11 - 2b$, which is an odd integer. But the only possibilities for the number of games of either colour played by either player are 4 or 5. It follows that Kim plays 5 games as White. Thus Kim starts by playing with the white pieces.
(Since $11 - 2b = 5$, we have $b = 3$. So to summarise, Kim wins 3 games as Black and 3 games as White, and Oli wins 2 games as Black and 1 game as White. Hence Kim wins the series 6 games to 3, and Black wins 5 games with White winning 4.)

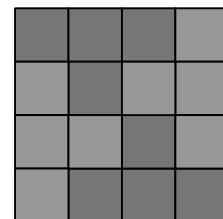
- M6.** The T-tetromino T is the shape made by joining four 1×1 squares edge to edge, as shown. The rectangle R has dimensions $2a \times 2b$, where a and b are integers. The expression ‘ R can be tiled by T ’ means that R can be covered exactly with identical copies of T without gaps or overlaps.



- (a) Prove that R can be tiled by T when both a and b are even.
 (b) Prove that R cannot be tiled by T when both a and b are odd.

Solution

- (a) Four tetrominos can be fitted together to form a 4×4 square, as shown alongside, and these can be tessellated inside a $4m \times 4n$ board.



- (b) Colour the board like a chessboard. Then a tetromino covers either three white squares and one black square, or one white and three black.

Let there be m of the first type of tetromino and n of the second. Then the total number of black squares they cover is $m + 3n$ and the total number of white squares is $3m + n$.

But the numbers of squares of each colour are equal, so $m + 3n = 3m + n$ and therefore $m = n$. Hence there are $4m$ squares of each colour, making $8m$ altogether.

It follows that $2a \times 2b = 8m$ and so $ab = 2m$. In other words, ab is even. Hence it is impossible to tile the board when both a and b are odd.

Alternative

Colour the board like a chessboard. Then a tetromino covers either one white square, or three white squares. So each tetromino covers an odd number of white squares.

Suppose that the board can be tiled. The board has $2a \times 2b = 4ab$ squares and each tetromino covers four of them, so there are ab tetrominos. Thus when both a and b are odd, there is an odd number of tetrominos.

Therefore, together the tetrominos cover an odd number of white squares—there is an odd number of them, and each of them covers an odd number of white squares. But the number of white squares is half the total number of squares, which is $2ab$, an even number.

Hence it is impossible to tile the board when both a and b are odd.